2021 - 2022 exam

① How many threads would you use for processing a million files?

→ as many as CPU cores. (usually)

→ if file processing involves I/O, use more threads than cores (because many threads will be blocked waiting for disk I/O, while others can run)

⇒ avoids excessive context switching.

CPU-bound → #threads ≃ #cores

I/O-bound → can use more threads than cores.

② See 12 jun 2026 (917)

③ Why do I/O operations cause a process to move from RUNNING state → WAIT state?

→ The process has to pause itself while it sends/gets data from I/O.

→ During this time it cannot continue execution, so it releases CPU and moves from RUN to WAIT until the I/O operation completes. (it leaves the processor for other processes)

ASC! ④ How is the address calculation done in the absolute fixed partition alloc?

ABSOLUTE FIXED PARTITION ALLOCATION: each partition has a fixed starting address known in advance.

~~↔ RELOCATABLE~~

physical address = partition start address + logical address

↳ offset inside the partition.

The program is loaded into a predetermined partition and uses absolute memory addresses directly, so no address translation is required

ex: partition start address: 1000
logical address: 50 (accessed by process)

2) phys. addr = 1000 + 50 = 1050

MEMORY ALLOCATION

I. FIXED partition allocation
- RAM is divided beforehand
  - ABSOLUTE FIXED
  - RELOCATABLE FIXED

II. DYNAMIC partition allocation
- Partitions created as needed
  - ⇒ external fragmentation?
    - First fit
    - Best fit
    - Worst fit
    - Next fit

III. PAGING (no external fragm.)
linear addr = base + offset
↓ paging
physical address

IV. SEGMENTATION (external fragm.)
linear addr = base + offset

V. SEGMENTATION + PAGING (nowadays)

⑤ ⊕ and ⊖ of the First-fit placement policy vs Worst-fit.

First-fit: ⊕ faster because it stops at the first block large enough for the request. (searches less memory)

⊖ may increase external fragmentation because of the many small unusable holes near the beginning of the memory. (wasted free space)

⑥ The most prioritary memory page that the NRU replacement policy chooses as victim page?

NRU = not recently used.

=> classifies pages using: (R, M)
- R → referenced bit
- M → modified bit

Classes: Class 0: (0,0) - not referenced, not modif. page
Class 1: (0,1) - not ref., modif. page
Class 2: (1,0) - ref., not modif. page
Class 3: (1,1) - ref., modif. page.

=> NRU chooses a victim from the lowest-numbered non-empty class.
↓
class 0.

⑦ size of block = B ; size of address = A ;
How many blocks are addressed by the triple indirect addressing of an i-node?

A block can store: $\frac{B}{A}$ addresses.

triple indirect => $\left(\frac{B}{A}\right)^3$

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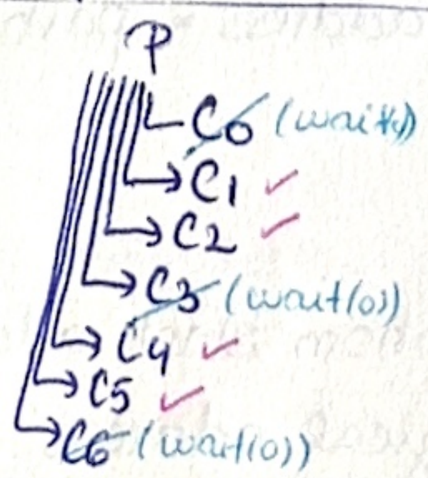


Right before calling the last wait(0), there were 5 active processes => MAX.

⑧ REGEX that accepts lines that contain "a" but don't contain "b".

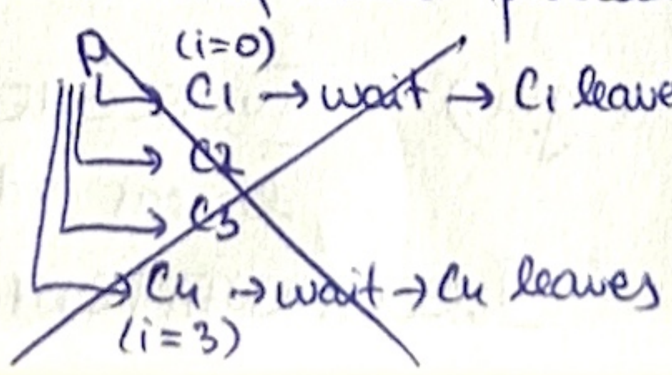
```
grep -E '^[^b]*a[^b]*$' filename
```

- start line ←
- [^b]* → 0 or more characters ≠ b
- must contain an "a"

⑨
```
for(i=0; i<7; i++){
    if(fork()==0){
        sleep(rand()%10);
        exit(0);
    }
    if(i%3==0){wait(0);}
}
```

pause ←

→ Max. nr. of child processes? (that can coexist)



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2